LPO1 – REFLECTION POSITIVITY ON THE INFINITE LATTICE: EXPANDED PROOF OUTLINE

Abstract. Starting from a coordinate infinite-volume limit of chamber QED3, this outline records the additional finite-volume and comparison arguments for reflection positivity at the six diagonal families. Heat-regularized local estimates, two-layer reduction and convex separation lead back to the original limiting law.

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1. Introduction and target statement

This is the companion outline to LPM1, synchronized with its referee revision of 20 October 2026. The preserved QEM1/S2 citations retain their historical locators; current LPM1 references identify the revised fold proof and Gaussian optimality argument. The outline takes the coordinate lattice limit as input; LPM1 also proves that construction. No continuum limit is taken here.

A. The infinite-lattice reflection positivity theorem

Model-specific. Overview Target.

Fix the lattice spacing and parameters $q > 0$, $0 < \rho < 1$, $M \in \mathbb{R}$ and $w_0, \dots, w_3 > 0$. Let μ_n be the curvature law of the all-flux chamber-fermion model on $(\mathbb{Z}/n\mathbb{Z})^3$, with antiperiodic fermions in each coordinate direction. The finite model and the results already available from its coordinate construction are specified in I; I.a fixes the field and reflection conventions.

Target theorem. For every integer sequence $n_i \geq 4$ tending to infinity, the periodic extensions of μ_{n_i} are tight. Every weak limit ν is invariant and reflection positive at every lattice mirror

$$x_j = c \quad (c \in \tfrac{1}{2}\mathbb{Z}), \qquad x_j \pm x_k = c \quad (c \in \mathbb{Z},\ j < k).$$

For each such mirror Π, let θ be its reflection and let Σ_+ be generated by plaquettes wholly in either chosen open side. Then the precise positivity assertion is

$$\int \overline{A(\theta^* F)} A(F)\, d\nu(F) \geq 0 \qquad (A \in L^2(\Sigma_+, \nu)).$$

These are three coordinate and six diagonal families. The law ν in this statement is the limit of the original cubic laws.

Proof route. I supplies tightness, coordinate RP and cubic symmetries. F compares a cubic expectation to a tilted law on a diagonal torus. N–E control that law and recover the coordinate link reflections after removing the tilt. C constructs its ordered limits. S transfers diagonal RP to ν by convex separation. For arbitrary original sides, V supplies the comparison on nearby divisible sides; for the divisible core, F.b already suffices.

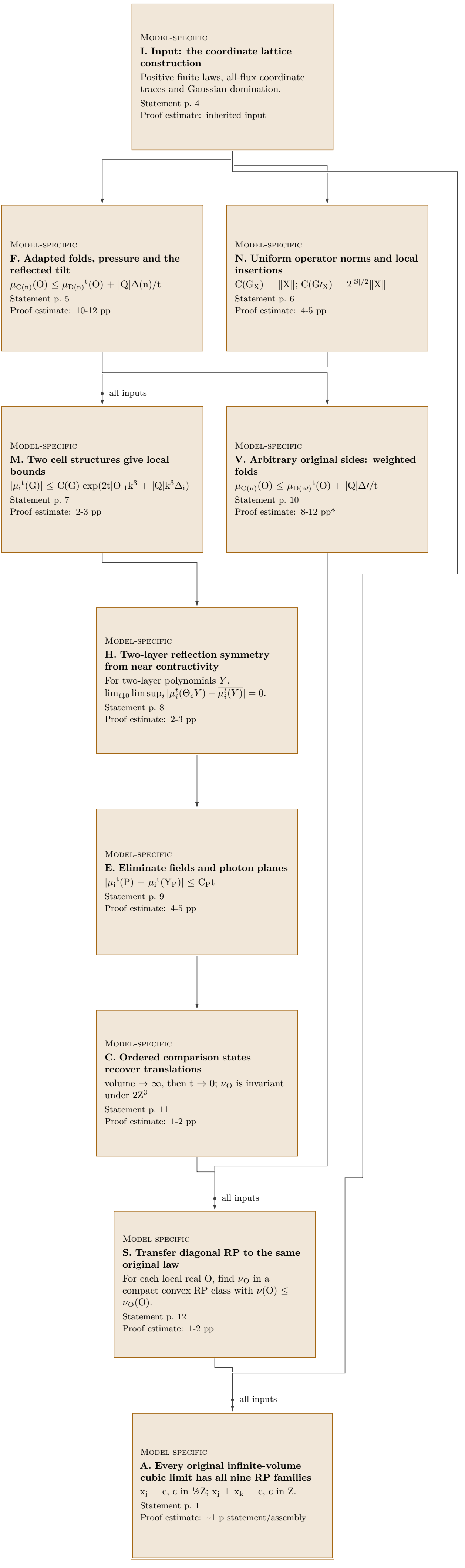
Reading and length. Each parent box opens a local map; each child opens a statement and its proof tasks. General results are identified explicitly. Other cards state the particular finite-model verification required. Budgets estimate a full proof, excluding linked inputs: about 25–33 pages for the divisible core, plus 8–12 for the arbitrary-side extension. They do not prescribe the length of these outline cards. The result asserts subsequential existence and positivity at fixed parameters. It makes no claim of uniqueness or of a continuum limit.

Source. Current companion: LPM1, Theorem A, referee revision of 20 October 2026. The statement and dependency structure are unchanged. The historical citations below refer to the preserved QEM1/S2 inputs: QED3, Theorem 4.7, p. 49, TeX 4130–4189 (divisible core). Retained S2, `cor:all-cubic-tori`, TeX 35529–35553 (all sides). The 263-page QED3 PDF omits the latter extension.

LPO1 – Reflection positivity on the infinite lattice: expanded proof outline

Definitions/setup | Model-specific | Interface (general)

Full hypotheses, source locations and proof estimates appear on the linked statement pages.



Arrows record mathematical dependencies; no distinction between proof and application inputs is made here. Every incoming branch at a junction is required together with the standing hypotheses on the statement page. Click a box for its statement. Return links are navigation, not proof dependencies.

The comparison entering S is V for arbitrary original sides, or F.b for the divisible core. The target for arbitrary sides uses V.

I. Input from the coordinate lattice construction

Model-specific. Overview Target.

Inherited input; its proof is the earlier coordinate lattice-limit packet. QED3 pp. 2–17, 26–30 and the coordinate passage on p. 49.

The new work begins with diagonal slab operators, not with the construction of the original finite probability measures.

Model-specific

I.a. Field space and the inherited probability theorem

The original cubic limits already have coordinate RP.

Statement p. 13

Reading order: The second input concerns finite fermion integrals.

Model-specific

I.b. Finite local insertion identities

Gauge covariance, local lifts, crossing modes and thin layers.

Statement p. 14

Definitions/setup Model-specific Interface (general)

Required finite-volume identities. On coordinate rectangular tori, retain positive normalizations, the all-flux slice trace and the sharp bound

$$\mu_{C(a,b,c)}(e^{\langle\beta,F\rangle}) \le e^{\|\beta\|^2/2}.$$

Here β is a torus cochain and its norm is the torus counting norm. Also retain $e^{-c_-|C|} \le Z_C[1] \le e^{c_+|C|}$ with finite fixed-parameter constants, and the common pressure ceiling on all comparison tori given in V.b. On all tori used below, retain the unnormalized finite functional $Z_T[G]$, its covariance under transport to the transformed torus, and its local insertion class. The adapted diagonal trace and the two folds are proved in F.a; they are not consequences of coordinate RP alone.

What needs to be read. The probability argument uses only I.a. The operator and elimination arguments also use I.b. Each result below identifies the finite identities it uses.

Source. QED3 §2, especially TeX 333–1329; coordinate trace and Gaussian domination, TeX 2124–2501. Earlier linked packet: coordinate lattice-limit outline.

3. Adapted folds and the reflected tilt

F. Adapted folds and the reflected tilt

Model-specific. Overview Target.

Full proof: about 10–12 pages beyond I; QED3 pp. 17–26 and 30–33.

Use periods ae_j for $C(a,b,c)$ and periods ae_-, be_+, ce_3 for $D(a,b,c)$, where $e_\pm = e_1 \pm e_2$. All fermion characters are -1; D' changes the be_+ character to $+1$. Write $D_n = D(n/2, n, n)$, of volume n^3.

Model-specific

F.a. Adapted slab traces and both cell folds

All fluxes, local lifts, adjoints and Schatten estimates.

Statement p. 15

Constant insertions give pressure control; bounded insertions give a local comparison.

↓

Model-specific

F.b. Pressure error and comparison of expectations

$\mu_{C(n)}(O) \le \mu^t_{D_n}(O) + |Q|\Delta(n)/t.$

Statement p. 16

Reading order: The following criterion is independent of the pressure comparison. Its application uses a full reflected array.

Interface

F.c. A factorized tilt preserves RP

A general lemma, applied to the adapted finite law.

Statement p. 17

Definitions/setup Model-specific Interface (general)

Output. For an observable inside an admissible diagonal cell, the comparison law has the desired wall RP at each positive strength. The comparison error vanishes at fixed strength as the volume grows. The tilt is extensive; its pointwise bound does not give uniform local estimates as volume grows. Those estimates are proved in M.

Source. QED3 §§3.1–3.4, 3.7 and 4.1, TeX 1452–2123 and 2503–2714.

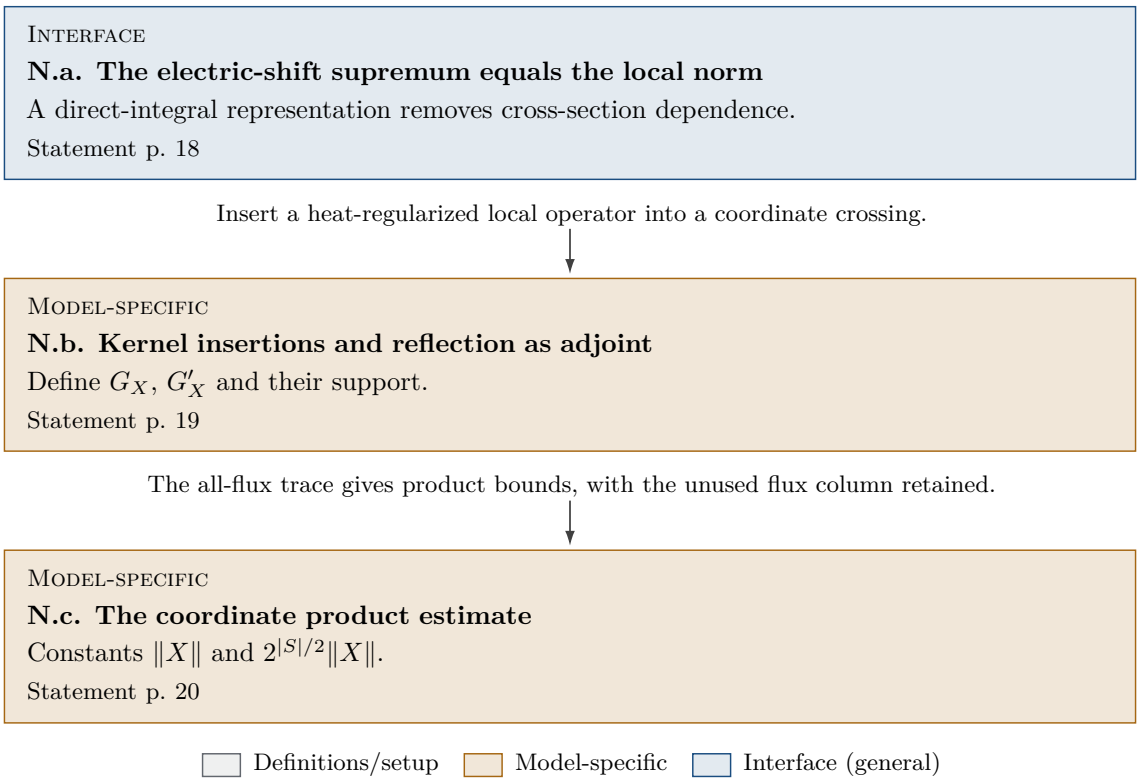
4. Operator insertions with a uniform coordinate bound

N. Operator insertions with a uniform coordinate bound

Model-specific. Overview Target.

Full proof: about 4–5 pages; QED3 pp. 33–38.

The amplification argument needs bounds for products of separated insertions, uniform in the cross-section and in every electric character. An estimate for a single finite-volume expectation would not suffice.



Output. For disjoint coordinate images of G_X, even with an arbitrary curvature character multiplying the whole product, the normalized finite functional is bounded by the product of local operator norms. For G'_X there is the explicit heat-variance factor. This is the finite input of M.b and the error control in H.b.

Source. QED3 §§4.2–4.3, TeX 2715–3124.

5. Amplification in two cell structures

M. Amplification in two cell structures

Model-specific. Overview Target.

Full proof: about 2–3 pages; QED3 pp. 38–40.

Fix a real cell observable O, support radius r_O, even width $h > 8r_O + 8$, and an admissible side sequence m_i. Write $|O|_1$ for its character coefficient norm, $\Delta_i = \Delta(m_i) \to 0$, and μ_i^t for the reflected tilt on D_{m_i}. The side sequence is eventually divisible by eight and has an infinite set $\mathcal{D} = \{d \in \mathbb{N} : d \mid m_i \text{ eventually}\}$, with $4h \in \mathcal{D}$. This full set is closed under divisors and least common multiples.

Interface

M.a. Separate every diagonal image into a coordinate cell

Injective placement, avoided tilt supports and free flux columns.

Statement p. 21

A diagonal fold followed by a coordinate fold turns a product estimate into a local bound.

Model-specific

M.b. Amplification and exponential moments

$|\mu_i^t(G)| \le C_G \gamma_{t,i}$ and tilted laws are tight.

Statement p. 22

Definitions/setup Model-specific Interface (general)

Main estimate. For each fixed support radius choose k, independently of the location, the strength and the volume, so that

$$\gamma_{t,i} = \exp(2t|O|_1 k^3 + |Q|k^3\Delta_i), \qquad |\mu_i^t(G)| \le C_G \gamma_{t,i}.$$

The same geometry gives $\mu_i^t(e^{\langle \beta, F\rangle}) \le e^{\|\beta\|^2/2}\gamma_{t,i}$. The order $i \to \infty$ then $t \downarrow 0$ makes the loss tend to one.

Source. QED3 §4.4, TeX 3125–3318.

H. Two adjacent layers become reflection symmetric

Model-specific. Overview Target.

Full proof: about 2–3 pages; QED3 pp. 40–42.

The near-isometric local bounds yield a general estimate for unital functionals. Applying it to the physical two-layer polynomials requires two different heat regularizations.

Interface

H.a. A nearly contractive unital functional is nearly Hermitian

An operator-algebra theorem with no lattice input.

Statement p. 23

Approximate a two-layer polynomial in a weaker heat norm while keeping the reflection operator bounded.

↓

Model-specific

H.b. Two-layer heat approximation and the ordered error

Volume first, then strength, then heat regularization.

Statement p. 24

Definitions/setup | Model-specific | Interface (general)

Output. For every gauge invariant polynomial Y reading only the two adjacent crossing layers at a coordinate link plane,

$$\lim_{t\downarrow 0} \limsup_{i\to\infty} |\mu_i^t(\Theta_c Y) - \overline{\mu_i^t(Y)}| = 0.$$

This is a finite-functional estimate, not an assertion of a fermionic weak limit. E reduces general local polynomials to this class.

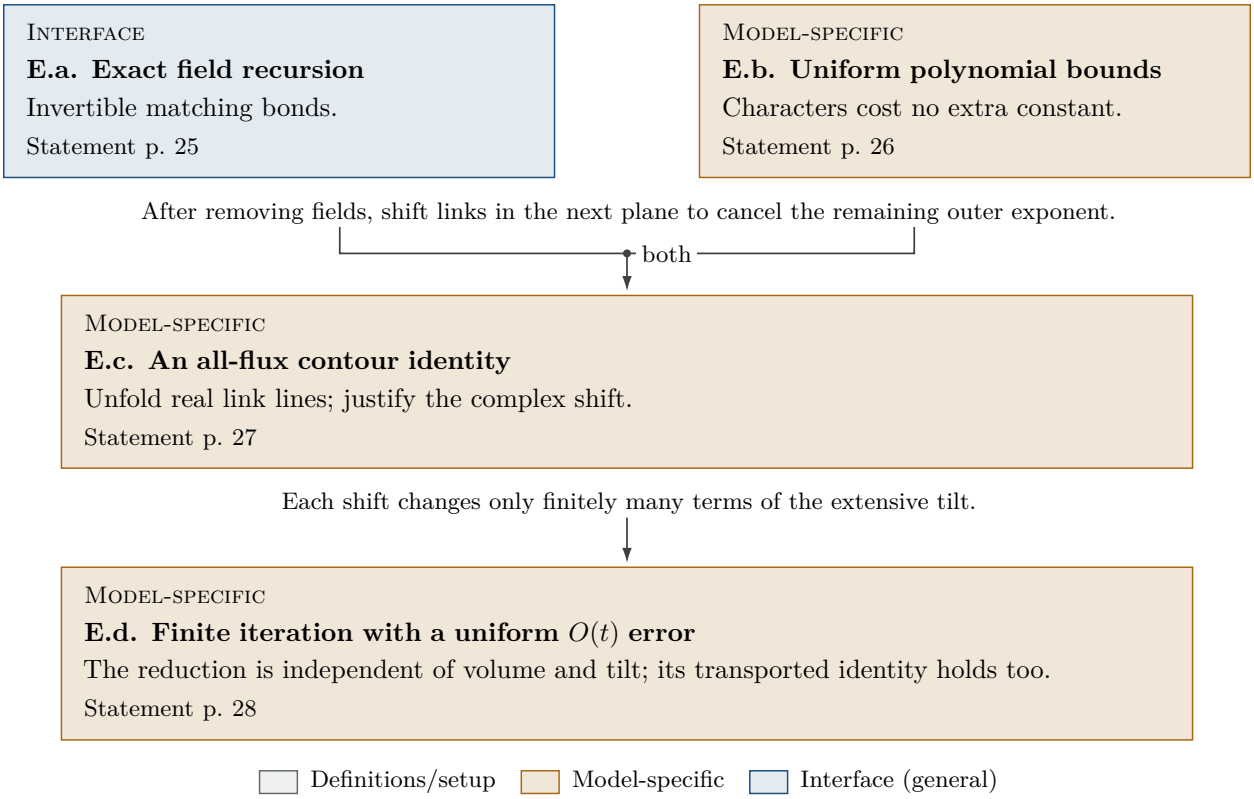
Source. QED3 §4.5, TeX 3319–3526.

E. Reduction to a prescribed pair of layers

Model-specific. Overview Target.

Full proof: about 4–5 pages; QED3 pp. 42–47.

The two-layer estimate does not apply to a general curvature character. Two finite operations remove its outer fields and photon exponents.



Output. For every local polynomial P of I.b and coordinate link plane c, there is a two-layer Y_P, independent of torus and strength, such that for each fixed lattice isometry g,

$$|\mu_i^t(P^g) - \mu_i^t((Y_P)^g)| \le C_{P,g}t \quad (0 \le t \le 1,\ i \ge i_0(P,g)).$$

Transport is proved on transformed tori; it cannot be deduced from a coordinate symmetry of D_{m_i}.

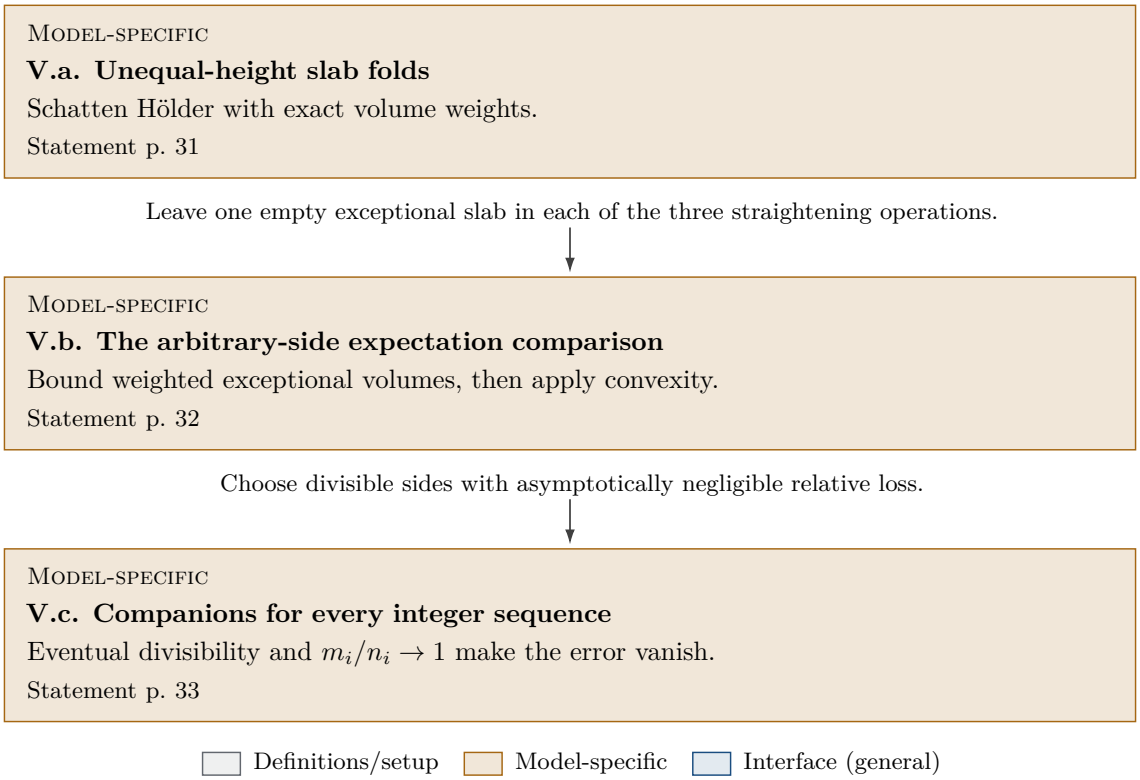
Source. QED3 §§4.6–4.7, TeX 3527–3941.

V. Arbitrary original sides via divisible companions

Model-specific. Overview Target.

Full proof: estimated 8–12 pages from retained S2, TeX 33984–35277. This branch is absent from the 263-page QED3 PDF.

This branch changes only the finite local comparison with the original cubic law. It uses the same comparison-state theorem C; the original cubic sequence is not replaced in the final theorem.



Output. For any $n_i \to \infty$, including odd sides, and each fixed cell observable O, there are admissible comparison sides m_i and errors $\Delta_i' \to 0$ such that

$$\mu_{C(n_i)}(O) \le \mu^t_{D_{m_i}}(O) + |Q|\Delta_i'/t.$$

Take the ordered limits supplied by C subordinate to the original convergent subsequence, then use S. The finite folded geometry may depend on O; the theorem quantifies over every original limit.

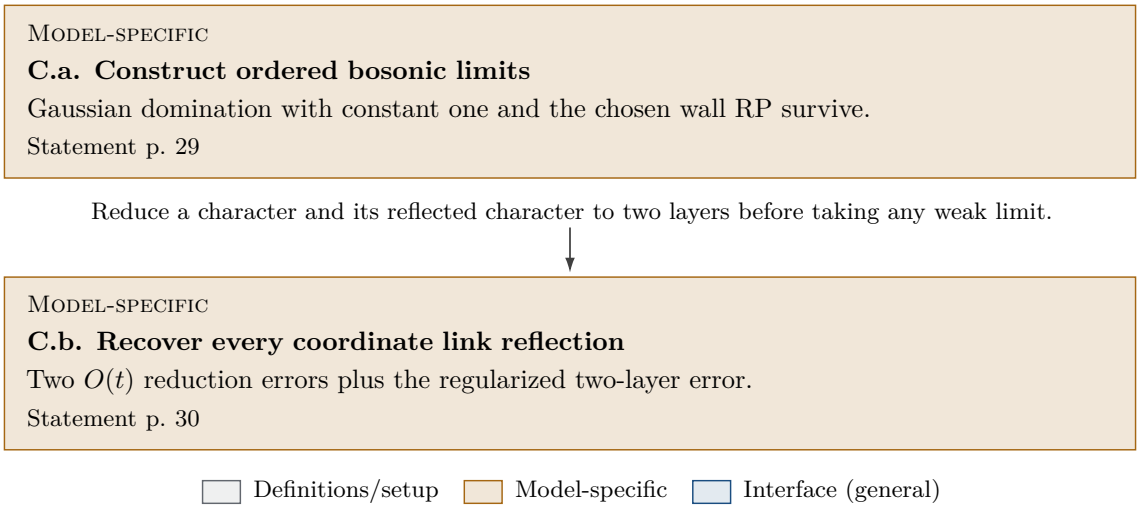
Source. S2 §§`Unequal slabs–Limits along every sequence`, TeX 33813–35553; the input order retains this lattice theorem.

C. Ordered comparison states and restored translations

Model-specific. Overview Target.

Full proof: about 1–2 pages; QED3 pp. 47–48.

The finite tilted laws have only wall-group periods. The symmetry needed to place arbitrary observables is a conclusion of the ordered limit, proved using finite-functional estimates before taking limits.



Output. Subordinate to any prescribed subsequence of comparison volumes there exists a law ν_O with Gaussian domination with constant one, RP and invariance at the chosen diagonal wall, and invariance under every coordinate link reflection. In particular it is invariant under $2\mathbb{Z}^3$ translations. Its construction depends on the cell observable O.

The local comparison with the original cubic law is used in S.c: F.b supplies it for the divisible core, and V for arbitrary original sides. The properties of the comparison state are the same in both applications.

Source. QED3 §4.8, TeX 3942–4041.

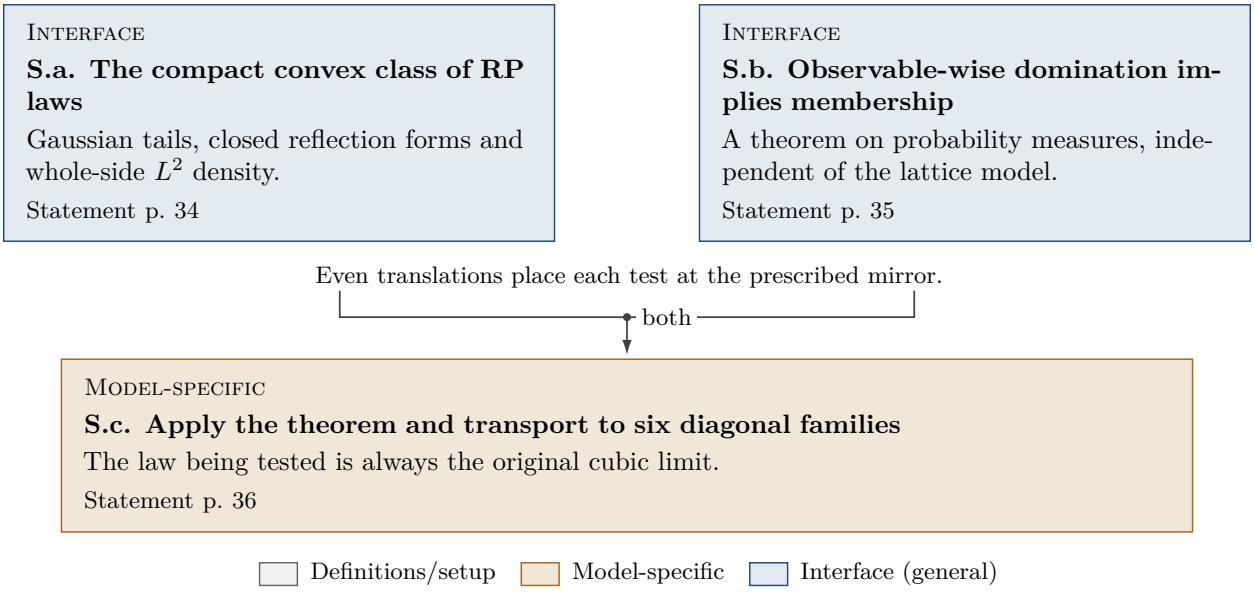
10. Compact convex transfer to the original limit

S. Compact convex transfer to the original limit

Model-specific. Overview Target.

Full proof: about 1–2 pages; QED3 pp. 48–49. The arbitrary-side application uses S2 TeX 35280–35553.

Comparison states depend on the observable. Convex separation allows that dependence while proving positivity for the same original law.



Output. For each integer c_0, the original limit belongs to the class with RP at $x_1 - x_2 = c_0$. Signed coordinate permutations give all six diagonal families. Combined with the coordinate input, this proves A. The support convention and L^2 conclusion are those of I.a.

Source. QED3 §4.9, TeX 4043–4189; retained S2 final lattice theorem.

Appendix A. Detailed proof tasks and supporting statements

The following pages expand the principal results. Each states its own inputs and gives the source location of the argument.

I.a. The field space and the probability input

Model-specific. Overview Parent I Target.

Let $\mathcal{P}$ be the countable set of positively oriented unit plaquettes of $\mathbb{Z}^3$. Set

$$\Omega = \{F \in \mathbb{R}^{\mathcal{P}} : dF = 0\}, \qquad \langle \beta, F\rangle = \sum_{\sigma} \beta_\sigma F_\sigma, \quad \|\beta\|^2 = \sum_{\sigma} \beta_\sigma^2.$$

The topology is the product topology; all tests β have finite support unless stated otherwise. Reverse plaquette orientation changes the sign of F. Pullback g^*F by a lattice isometry includes that sign. Write $\varphi_\beta(F) = e^{i\langle \beta, F\rangle}$. Real local trigonometric polynomials form the real span of the real and imaginary parts of these characters, including constants.

Inherited theorem. For fixed q, ρ, M, w_i as in A, the positive cubic laws μ_n, $n \geq 4$, have Gaussian domination for torus cochains with the torus counting norm, and are invariant under $\mathbb{Z}^3$ translations, signed coordinate permutations and $F \mapsto -F$. Their periodic extensions are tight, and every weak limit has these symmetries, Gaussian domination for every finite infinite-lattice cochain with its counting norm, and coordinate RP on the entire open-side L^2 space. A lifted finite-volume bound uses the unaliased norm only when the test's plaquettes remain distinct modulo periods. Each fixed finite test has this property for all large sides; single-plaquette tails hold for all $n \geq 4$.

Optimality for the original cubic limits. Choose a nonzero finite closed integer two-cochain z_0 and put $\beta = (2\pi/q)z_0$. On sufficiently large cubes, completing the square and reindexing the entire closed-cochain sum gives $\mu_n(e^{\langle \beta, F\rangle}) = e^{\|\beta\|^2/2}$. The identity at 2β gives uniform second moments, so truncation passes equality to every original cubic weak limit. This proves optimality of the common coefficient for all finite real sources; it asserts equality only for these specified sources.

Reflections and support. If $\Pi = \{\ell(x) = c\}$, where $\ell = x_j$ or $x_j \pm x_k$, set

$$\theta x = x - \frac{2(\ell(x) - c)}{|\nabla \ell|^2} \nabla \ell, \qquad (\Theta A)(F) = \overline{A(\theta^* F)}.$$

The positive-side sigma-algebra is generated by F_σ for which every corner of σ lies in $\{\ell > c\}$. In a finite volume its test must fit in the open positive half-period slab. No variable on or crossing the mirror is included in this definition.

Where the input is used. F.b uses positive normalization and translation invariance; M.b uses rectangular Gaussian domination; S.c uses translation invariance of the original limit and its signed permutation symmetry. The coordinate compactness proof is not repeated in these cards.

Source. LPM1 §5.3, including the unnumbered optimality argument added in the referee revision of 20 October 2026. Historical support for the coordinate construction: QED3 TeX 333–615, 2402–2501 and 4130–4171; pp. 2–8, 28–30 and 49. The optimality addition is not attributed to that frozen source.

I.b. Finite local data needed for the diagonal argument

MODEL-SPECIFIC. Overview Parent I Target.

Finite functional. Put $r_q = 2\pi/q$. Photon configurations are (A, z) with A in a link phase cell of width r_q and z a closed integer plaquette cochain, with all flux classes summed. Their curvature is $F = dA + r_q z$, and $(A, z) \sim (A + r_q k, z - dk)$. The functional $Z_T[G]$ integrates $e^{-\|F\|^2/2} e^{-\bar\psi \mathcal{D}_A \psi} G$ against this photon measure and the finite Berezin measure, with $\int e^{-\bar\psi \mathcal{D}_A \psi} = \det \mathcal{D}_A$. Only its bosonic normalization is a probability law.

Local class. For a box B, an insertion reads only its links and fields at $\mathcal{X}(B) = \{x : \pi(x) + [-1/2, 1/2]^3 \subset B\}$, where π is projection to the photon site. It is even and has finitely many Grassmann coefficients a_Ψ, satisfying $a_\Psi(\widehat{A} - d\lambda) e^{iq\langle \lambda, \mathfrak{q}(\Psi)\rangle} = a_\Psi(\widehat{A})$, each bounded by $C \exp((1/2 - \epsilon) \sum_{\sigma \subset B} F_\sigma^2)$, $\epsilon > 0$. It is read at a local lift $\widehat{A}$ with $d\widehat{A} = F$ and $e^{iq\widehat{A}} = e^{iqA}$. Individual translates of $B + [-1, 1]^3$ must be disjoint under the period group; different insertion boxes and all their period translates must also be disjoint. Bounded curvature functions are allowed.

The polynomial subclass used in E is

$$P(\widehat{A}) = \sum_a b_a e^{i\langle \delta_a, \widehat{A}\rangle} \Psi_a, \qquad d^* \delta_a = q\mathfrak{q}(\Psi_a).$$

Here the finite-support charge $\mathfrak{q}$ counts ψ minus $\bar\psi$ at their projected photon sites. This equality is gauge invariance and implies even parity.

Chamber data used later. A coordinate link crossing has positive mode weights and a finite Fock space. Reflection exchanges its two boundaries by the antilinear, order-reversing fermion reflection. Each thick coordinate layer contains six thin layers; successive layers are joined by a perfect matching. Its hopping matrices have eigenvalues $w_i(\rho \pm 1)/2$, hence are invertible. An intervening matching vertex has only one neighbor in the layer being removed. These are the graph facts used in E.a. They are verified for the chamber action in the cited source.

Source. QED3 TeX 365–563, 779–999, 1452–1534, 3319–3345 and 3527–3559; pp. 4–7, 10–13, 17–18 and 40–43. These identities are inherited from the finite-volume construction.

F.a. Slab operators and the two changes of frame

MODEL-SPECIFIC. Overview Parent F Target.

Inputs and statement. Use the finite functional and local insertion class of I.b. For a mirror height ℓ with normal n_ℓ, take a rank-two cross-section in $\{\ell = 0\}$ of systole at least four; slab endpoints lie in $(|n_\ell|^2/2)\mathbb{Z}$. Construct boundary Hilbert spaces and Hilbert–Schmidt slab operators $\mathfrak{S}[G]$ with exact gluing and reflection as adjoint. For a closing vector g of minimal positive height $p = \ell(g)$, so $\ell(\Gamma) = p\mathbb{Z}$, and character $\chi(g) \in \{\pm 1\}$, the trace is

$$Z_T[\prod G_j] = \mathrm{Tr}(V_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2} \mathfrak{S}[G_r] \cdots \mathfrak{S}[G_1]).$$

Here V_g is boundary transport and U_{f} is Fock parity; the closing factor is unitary. The straightened torus T^0 uses $g^0 = pn_\ell/|n_\ell|^2$ with character -1; require $g^0 \in \mathbb{Z}^3$ and its period lattice to have systole at least four. For $2k$ equal slabs and the torus T^0 with straight normal closing vector, this gives

$$Z_{T^0}[\mathrm{fold}\, G] = \|\mathfrak{S}[G]\|_{2k}^{2k}, \qquad |Z_T[\prod_{j=1}^{2k} G_j]| \le \prod_{j=1}^{2k} Z_{T^0}[\mathrm{fold}\, G_j]^{1/(2k)}.$$

For orthogonal cells let $\|G\|_T = Z_T[\mathrm{Fold}_T\, G]^{1/N}$, where N is their number. The root is real and nonnegative. Successive folds give $|Z_T[\prod G_\kappa]| \le \prod \|G_\kappa\|_T$ on a carrying rectangular torus. In the diagonal frame the fold from $C(n)$ or D'_n ends on D_n; in the coordinate frame the fold from $D(a, 2a, c)$ ends on $C(a, 4a, c)$. The requisite widths divide the corresponding half-periods; in particular $n \in 2h\mathbb{Z}$, $n \ge 8$ for the diagonal fold, and $a, c \in 2h\mathbb{Z}$, $a, c \ge 4$ for the other fold.

Proof tasks. First establish local-lift independence and coefficient integrability. For diagonal mirrors verify that crossing bonds are normal and join reflected endpoints; positive Clifford projections give their Gram expansion. Poisson summation and cut-link integration supply the nonnegative Gauss constraint. Retain mirror and cut variables, crossing modes and every integer flux when defining the kernels. Gaussian summation gives their Hilbert–Schmidt bounds; gluing integrates each cell exactly once. The general trace retains closing transport and Fock parity. For each selected slab $[a, a + h']$, keep the actual initial boundary $a = c_{j-1}$ and the map $Y = \mathfrak{S}[G_j] : \mathcal{H}(a) \to \mathcal{H}(a + h')$. Form the fold on $[a, a + p]$. The pair translation $v = 2h'n_\ell/|n_\ell|^2$ is integral, and the pair products telescope to $V_{kv}(Y^*Y)^k$. The antiperiodic closing vector $g^0 = kv$ cancels this transport without parity, leaving the trace on $\mathcal{H}(a)$. Do not identify different starting boundaries by a half-lattice translation. Only then apply Schatten Hölder to the cyclic trace.

Straightening successively gives

$$C(n) \longrightarrow D'_n \longrightarrow D_n \longrightarrow C(n/2, 2n, n).$$

Track the character as well as the period lattice at each arrow. For three-dimensional cell folds, the three slab exponents multiply to $1/N$. Check individual enlarged-box embedding and disjointness of all period translates throughout; open-cell support alone must be connected to these conditions by the source geometry.

Source. LPM1, Proposition 4.2, proof at the actual starting boundary; the supplied argument is S2, TeX 18004–18052. This corrects the starting-slab translation in the historical QED3 proof. Historical kernel and comparison material: QED3 pp. 17–25, TeX 1452–2045; slab kernels, trace, `prop:slab-fold-holder` and `thm:slab-chessboard`.

F.b. Pressure loss becomes a local expectation error

MODEL-SPECIFIC. Overview Parent F Target.

Setup. Use F.a, positive exponential partition bounds from I, and translation invariant cubic laws. Define $f(a,b,c) = (abc)^{-1} \log Z_{C(a,b,c)}[1]$. For even $n \geq 8$, the four partition functions in the chain of F.a have equal volume and increase from left to right. The coordinate half-step trace $Z_{C(a,b,c)}[1] = \operatorname{Tr}(Y^*Y)^a$ implies that f decreases separately in each side. Therefore

$$f \to f_\infty \in \mathbb{R}, \qquad 0 \leq \Delta(n) := f(n/2, 2n, n) - f(n,n,n) \to 0.$$

Their logarithmic normalization losses are at most $n^3\Delta(n)$.

Local comparison theorem. Let Q be an open diagonal cell of even height width h and volume $|Q| = h^3/2$, and let O be a real local trigonometric polynomial strictly inside it. If $4h \mid n$, form V_n by summing the reflected copies of O in all $N = n^3/|Q|$ cells of D_n. Set $\mu^t_{D_n}(G) = Z_{D_n}[e^{tV_n}G]/Z_{D_n}[e^{tV_n}]$. Then

$$\mu_{C(n)}(O) \leq \mu^t_{D_n}(O) + \frac{|Q|\Delta(n)}{t} \qquad (t > 0).$$

Proof tasks. On $C(n)$, sum translated copies $\widetilde{V}_n$ rather than reflected ones. Jensen gives $\log \mu_{C(n)}(e^{t\widetilde{V}_n}) \geq tN\mu_{C(n)}(O)$. The diagonal chessboard estimate compares its unnormalized partition function to $Z_{D_n}[e^{tV_n}]$. If $L(t) = \log \mu_{D_n}(e^{tV_n})$, convexity gives $L(t) \leq tL'(t)$. Wall-group invariance makes $L'(t) = N\mu^t_{D_n}(O)$. The remaining logarithmic normalization difference is at most $n^3\Delta(n)$; divide by tN.

The bound controls pressure per volume. It does not bound the partition ratio uniformly. The factor $1/t$ requires the volume limit first.

Source. QED3 pp. 25–26 and 30–31, TeX 2047–2123 and 2503–2596; `prop:slab-pressure`, `prop:slab-finite-fold`.

F.c. Factorized perturbations preserve reflection positivity

INTERFACE. Overview Parent F Target.

General theorem. Let μ be a reflection invariant probability law, with involution θ exchanging two side sigma-algebras. Suppose $\mu((\Theta A)A) \geq 0$ for the positive-side tests under consideration, where $\Theta A = \overline{A \circ \theta}$. Let h be bounded and positive-side measurable, and suppose $w = (\Theta h)h$ is real, nonnegative and $0 < \mu(w) < \infty$. Then $d\mu_w = w\, d\mu/\mu(w)$ is reflection invariant and RP, provided the original test class contains hA. Indeed its reflection form is $\mu((\Theta(hA))(hA))/\mu(w)$. The same identity applies to the positive Gram form of finite even Grassmann insertions; it does not turn a Grassmann functional into a probability measure.

Application to the reflected array. Fix offsets $c_-, c_+ \in \mathbb{Z}$, $c_3 \in \mathbb{Z} + 1/2$, and an even width h. Cells are bounded by consecutive levels of $(\ell_-, \ell_+, x_3) = (x_1 - x_2, x_1 + x_2, x_3)$, distance h in each height. Assume O is strictly inside one cell, and transport it by successive wall reflections to define V_i on D_{m_i}, with $4h \mid m_i$ eventually. Thus each height period has an even number of cells. For each cell wall, half the cells lie in either open half-period slab. If $h_+ = \exp(t \sum_{Q_\kappa \text{ in positive slab}} O^{[\kappa]})$, then exactly

$$e^{tV_i} = (\Theta h_+)h_+.$$

The finite adapted Gram form of F.a therefore gives wall RP of μ_i^t. Its bosonic law is invariant under the wall group; in particular under translations by he_-, he_+ and $2he_3$.

Checks and output. Strict open-cell support makes h_+ an allowed side insertion. Boundedness gives a positive denominator, with $Z_{D_{m_i}}[e^{tV_i}] \geq e^{-tN_i|O|_1} Z_{D_{m_i}}[1]$, where $|O|_1$ is the sum of absolute character coefficients. The conclusion is RP at the cell walls and the stated wall periods. All coordinate link reflections are recovered only in C.b.

Source. QED3 pp. 32–33, TeX 2623–2714, especially the factorizations 2663–2679. The general statement records the identity used there.

N.a. A local norm uniform over quotient realizations

INTERFACE. Overview Parent N Target.

Objects. Let S be finitely many spatial links, Λ a finite set containing their endpoints, and $\mathcal{F}_\Lambda$ the finite charged crossing Fock space. On $H_S = L^2(\mathbb{R}^S) \otimes \mathcal{F}_\Lambda$, set $E = -i\partial_A$. Consider finite sums

$$X = \sum_\delta e^{i\delta A} \Upsilon_\delta(E), \qquad d_s^* \delta = q z(\delta), \quad z(\delta) \in \mathbb{Z}^\Lambda.$$

Each symbol is a finite sum of $e^{-Q(\xi_{S'})+\lambda\xi_{S'}}$ times a matrix of Fock charge $z(\delta)$, meaning $u(a)\Upsilon u(a)^{-1} = e^{iq\langle a, z(\delta)\rangle}\Upsilon$; u is the diagonal gauge unitary on the charged Fock space. Q is positive definite on $S' \subset S$. The empty subset permits constants. This is a unital $*$-algebra of bounded gauge invariant operators. Let $\alpha_\beta X = e^{i\beta A} X e^{-i\beta A}$. For a cross-section Ξ with rectangular slice Σ, let d_s be its coboundary and d_s^* its counting adjoint. Put $\Gamma_s = r_q(\mathbb{Z}^{\Sigma^{(1)}} \cap \ker d_s)$, $r_q = 2\pi/q$, and use $L^2(\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s)$ with quotient Lebesgue measure, tensored with the full crossing Fock space. Require the local support hull to embed in Σ. In this space, X_Ξ is the same displayed sum, using quotient link multipliers and derivatives and the tensor inclusion of local Fock matrices, with their crossing-order signs.

Theorem. For each such Ξ,

$$\sup_{\beta \in \mathbb{R}^S} \|(\alpha_\beta X)_\Xi\| = \|X\|_{\mathcal{L}(H_S)}.$$

Consequently the supremum over all Ξ, β has this same norm. Enlarging S or Λ preserves it. An individual unshifted quotient norm need not equal the local norm.

Proof tasks. The charge condition makes the phase well defined on the quotient; use the integer box Poincaré argument on the support hull. The Fock inclusion is an injective $*$-homomorphism, hence isometric. Choose a Borel representative $A^\flat$ of $[A]$ in the quotient. Adjoin a real variable $z \in \mathbb{R}^S$. The two changes $z \mapsto z + A^\flat|_S$ and $[A] \mapsto [A - z]$ conjugate the local operator to $\sum e^{i\delta A}\Upsilon_\delta(E + E_z)$. Fourier transformation in z identifies it with the direct integral of $(\alpha_{-p}X)_\Xi$. Uniform continuity of the symbols makes this family norm continuous, so its essential supremum equals its ordinary supremum. This proves the equality.

Source. QED3 pp. 33–35, TeX 2715–2852; `prop:diag-local-norm`. This is an operator theorem for the specified charged quotient realizations; the charge and symbol hypotheses are part of the statement.

N.b. Heat kernels inserted at a crossing

MODEL-SPECIFIC. Overview Parent N Target.

Inputs and definition. Use the positive crossing expansion of I.b and an operator X from N.a. On a spatial slice write A_j for its spatial links and a_j for the temporal links to slice $j+1$. Then $F_j^{\mathrm{el}} = \widehat{A}_{j+1} - \widehat{A}_j - d_s\widehat{a}_j$ on a local lift. Choose ordered crossing modes ζ_i with charges $Q_i = \pm 1$ at spatial projected sites y_i. Put $\zeta_I = \prod_{i\in I}\zeta_i$, $\mathfrak{q}(I) = \sum_{i\in I} Q_i 1_{y_i}$, and $u(a)e_I = e^{iq\langle a,\mathfrak{q}(I)\rangle}e_I$. For a matrix K, define

$$\mathrm{Ins}_c[K] = \sum_{I,J} K_{IJ}(\Theta_c\zeta_J)\zeta_I, \qquad e^{-S_c^{\times}} = \mathrm{Ins}_c[\omega u(a_j)].$$

Here $\omega e_I = (\prod_{i\in I}\omega_i)e_I$ with $\omega_i > 0$; these identities specify the finite crossing basis and its weights. For $s = 1/4$ or $1/8$ let $k_X^{(s)}$ be the kernel of $e^{-s|E|^2}Xe^{-s|E|^2}$. At the link plane $c = j + 1/2$, insert this kernel in place of the electric Gaussian and fermion crossing factor:

$$G[k] = \mathrm{Ins}_c[\omega^{1/2}k(\widehat{A}_{j+1}, \widehat{A}_j + d_s\widehat{a}_j)u(\widehat{a}_j)\omega^{1/2}] \prod_{b\in S} g_1(F_j^{\mathrm{el}}(b))^{-1} e^{S_c^{\times}(\Lambda)}.$$

Here $g_1(x) = (2\pi)^{-1/2}e^{-x^2/2}$; Ins_c is the finite matrix-to-crossing-mode expansion, and $S_c^{\times}$ is the crossing part of the fermion action. Set $G_X = G[k_X^{(1/4)}]$, $G'_X = G[k_X^{(1/8)}]$.

Statement. These are torus-independent, even, gauge invariant local insertions. They satisfy the subcritical coefficient bound of I.b,

$$G_1 = 1, \qquad \Theta_c G_X = G_{X^*}, \qquad \Theta_c G'_X = G'_{X^*}.$$

Coordinate translations and spatial reflections transport them to operators with the same norm. Use the covariantly chosen box $B^{\#}$ obtained by doubling the largest half-width of the rectangle containing both spatial copies of S, the temporal links over Λ and the unit cubes at their two endpoint slices.

Proof tasks. Compute the regularized kernel of one Gaussian symbol by Fourier transform. Its difference variable has strict Gaussian decay. Gauge covariance of the kernel gives gauge invariance of $G[k]$; the canceled electric weight leaves growth exponent strictly below $1/2$. Exchanging its two arguments and taking adjoints proves the link-reflection identities. For spatial reflections, include the crossing-mode reordering signs; the resulting antiunitary conjugation preserves both the charge condition and the heat factors.

Source. QED3 pp. 35–37, TeX 2854–2981; kernel, support and reflection formulas 2859–2981.

N.c. A coordinate trace bound for disjoint insertions

MODEL-SPECIFIC. Overview Parent N Target.

Statement. Let G be an insertion supported in a box B. Say it has coordinate bound C_G if on every rectangular coordinate torus T,

$$\left| Z_T[\varphi_\beta \prod_{a=1}^{r} G^{g_a}] \right| \le C_G^r Z_T[1]$$

for every character and every nonempty family of translation and signed coordinate-reflection images. Require individual enlarged-box embedding, disjointness of the whole family and its period translates, and, in each coordinate direction μ, a plaquette σ orthogonal to e_μ such that none of those boxes or translates contains any plaquette $\sigma + ke_\mu$. All periods have systole at least four.

For the insertions of N.b, the following constants work:

$$C_{G_X} = \|X\|, \qquad C_{G'_X} = 2^{|S|/2}\|X\|.$$

A bounded bosonic insertion has $C_G = \|G\|_\infty$. The bounds persist under box enlargement, multiplication by a curvature character, and signed coordinate permutations.

Proof tasks. Choose the magnetic flux representative on the unread column. This makes the unfolded path variables genuine local lifts in every insertion hull. Cancel the original crossing kernels there and insert the regularized ones. Strict Gaussian decay gives absolute integrability and Hilbert–Schmidt factors before any cyclic trace manipulation.

An electric character changes E to $E - \beta$ and X to $\alpha_\beta X$. The norm theorem N.a bounds every local middle operator by $\|X\|$. In the cyclic product apply Schatten Hölder with equal exponents $2n_1$. The resulting positive traces have the original bosonic law. The $1/8$ heat factors replace g_1 by $g_{1/2}$ on the selected links; positivity bounds this change by $2^{\sum_a |S_a|/2}$. The $1/4$ case has no variance loss.

The free-column assumption is needed to identify local lifts in the all-flux trace. M.a supplies it for the placements actually used.

Source. QED3 pp. 37–38, TeX 2982–3124; `prop:diag-coordinate-insertions`, including its trace proof.

M.a. The geometric placement for the second fold

INTERFACE. Overview Parent M Target.

Inputs and claim. Use the diagonal array of M, and a support box $B = y + [-r, r]^3$. Choose $h'' \in h\mathbb{Z}$ and $k \in \mathbb{N}$ with

$$h'' \geq 2r + 6h, \quad kh \geq 16h'', \quad kh \in h''\mathbb{Z}, \quad 4h'', 2kh \in \mathcal{D}.$$

There are coarse diagonal offsets, differing from the original offsets by multiples of h, and half-integer coordinate offsets of width h'', such that every coarse diagonal image gB lies strictly inside a different coordinate cell, also modulo the period lattice. The coordinate walls avoid every reflected copy of $\operatorname{supp} O$. All enlarged boxes embed, and their coordinate folds have the unread columns required by N.c.

Proof tasks. Put the center heights of B, relative to the three lower coarse walls, in $[2h'', 2h'' + h)$, $[4h'', 4h'' + h)$ and $[h'', h'' + h)$. For a coarse wall-group element with reflection labels $a \in \{0, 1\}^3$ and translation labels $z \in \mathbb{Z}^3$, write

$$gy = y + h''\kappa(g) + s(a), \qquad |s(a)|_\infty < 2h,$$

$$\kappa(g) = (-2a_- - 4a_+,\, 2a_- - 4a_+,\, -2a_3) + \frac{kh}{h''}(z_- + z_+, z_+ - z_-, 2z_3).$$

The eight first vectors are distinct with differences at most six; $kh/h'' \geq 16$ makes κ injective. The period translations change its value by the period divided by h'', proving injectivity modulo periods as well.

The admissible offset interval in each coordinate has length at least $2h$. Reflected tilt supports forbid at most $8r_O + 4$ half-integer residue classes modulo h, so $h > 8r_O + 8$ leaves a choice. The period and support margins give enlarged-box embedding. Choose a plaquette column whose every plaquette meets a transverse half-integer coordinate wall. No such plaquette is contained in any insertion box inside an open cell; this supplies the free columns. The two cell counts are $N_i' = m_i^3/(|Q|k^3)$ and $N_i'' = m_i^3/(h'')^3$.

Source. QED3 pp. 38–39, TeX 3125–3217. This finite placement lemma is specific to the two stated frames.

M.b. Local amplification and tightness of the tilted laws

MODEL-SPECIFIC. Overview Parent M Target.

Theorem. Use the finite folds of F.a, the pressure error of F.b, and the placement of M.a. If G has coordinate bound C_G in the sense of N.c, then for all large i and all $t \geq 0$,

$$|\mu_i^t(G)| \leq C_G \gamma_{t,i}, \qquad \gamma_{t,i} = e^{2t|O|_1 k^3 + |Q| k^3 \Delta_i}.$$

The threshold and k depend on the fixed support radius and cell data, not its location or t. For a real cochain β in that box,

$$\mu_i^t(e^{\langle \beta, F\rangle}) \leq e^{\|\beta\|^2/2} \gamma_{t,i}.$$

Hence the bosonic laws are tight at each fixed t, and every volume limit ν_t has $\nu_t(e^{\langle \beta, F\rangle}) \leq e^{\|\beta\|^2/2 + 2t|O|_1 k^3}$.

Proof tasks. In one coarse diagonal cell put PG, and put P in all others, where P is the part of e^{tV_i} in that cell. The first fold gives

$$|\mu_i^t(G)|^{N_i'} \leq \frac{Z_{D_{m_i}}[e^{tV_i} \prod_{a=1}^{N_i'} G^{g_a}]}{Z_{D_{m_i}}[e^{tV_i}]}.$$

Expand the tilt in characters with total coefficient norm at most $e^{tN_i|O|_1}$. The placement lets the second, coordinate fold apply N.c to each occupied cell. Its numerator is bounded by $e^{tN_i|O|_1} C_G^{N_i'} Z_{C(m_i/2, 2m_i, m_i)}[1]$. The denominator contributes another $e^{tN_i|O|_1}$, and pressure costs $e^{m_i^3 \Delta_i}$. Taking the N_i'-th root gives the estimate.

For exponentials, replace the insertion bound by the rectangular Gaussian bound; disjoint copies have squared norm equal to their number times $\|\beta\|^2$. One-plaquette exponential tails yield product tightness. Truncated exponentials pass the displayed bound to weak limits. No limiting fermionic functional is required.

Source. QED3 pp. 39–40, TeX 3218–3318; `thm:diag-amplification` and the exponential-moment argument.

H.a. Near contractivity implies near Hermiticity

INTERFACE. Overview Parent H Target.

General theorem. Let $\mathfrak{A}$ be a unital $*$-subalgebra of bounded operators on a Hilbert space, with its operator norm, and let $o : \mathfrak{A} \to \mathbb{C}$ be linear, $o(1) = 1$, and $|o(X)| \le \gamma \|X\|$ for all X, where $\gamma \ge 1$. Then

$$|o(X^*) - \overline{o(X)}| \le 4\sqrt{\gamma^2 - 1}\,\|X\|.$$

No representation by a positive measure is assumed.

Proof. For $X = X^*$ and $\|X\| \le 1$, write $o(X) = u + iv$. For real s,

$$(v + s)^2 \le |o(X + is)|^2 \le \gamma^2 \|X + is\|^2 \le \gamma^2(1 + s^2).$$

If $\gamma > 1$, choosing $s = v/(\gamma^2 - 1)$ gives $|v| \le \sqrt{\gamma^2 - 1}$; if $\gamma = 1$, let s tend to either infinity. Apply this to the real and imaginary self-adjoint parts of X to obtain the estimate.

Application and uniformity. For the local operator algebra of N.a, $o_i^t(X) = \mu_i^t(G_X)$ is unital by N.b. The amplification theorem M.b gives the required bound with $\gamma = \gamma_{t,i}(k)$. When applying the general estimate to a fixed X, choose one support box containing X, X^* and 1, so the same k works on the subalgebra with that support. Reflection as adjoint then identifies the left side with the reflection defect of G_X.

Source. QED3 p. 42, TeX 3481–3508. The general proof above is exactly the operator-functional argument supplied there.

H.b. Two heat scales and the two-layer reflection estimate

MODEL-SPECIFIC. Overview Parent H Target.

Objects and inputs. A two-layer polynomial Y at $c = j + 1/2$ reads spatial links in planes $j, j+1$, their connecting temporal links, and fields only at the crossing endpoints. Use I.b, N.b–N.c, M.b and H.a. A monomial has spatial exponents δ_0, δ_1 and a finite matrix Υ_Y obtained from the crossing-mode basis. Gauge invariance fixes its temporal exponent. Its kernel is

$$k_Y^0(A', A) = e^{i\delta_1 A'} g_1(A' - A) e^{i\delta_0 A} \Upsilon_Y, \qquad Y = G[k_Y^0].$$

Products of g_1 over the spatial link set S are understood.

Approximation theorem. There is a bounded X'_Y in the local algebra with $k_Y^0 = e^{-|E|^2/8} X'_Y e^{-|E|^2/8}$. For $P_\epsilon = e^{-\epsilon|E|^2}$ there is a bounded $X_{Y,\epsilon}$ with

$$e^{-|E|^2/4} X_{Y,\epsilon} e^{-|E|^2/4} = P_\epsilon k_Y^0 P_\epsilon,$$

$$Y - G_{X_{Y,\epsilon}} = G'_{X'_Y - P_\epsilon X'_Y P_\epsilon}, \qquad \xi_Y(\epsilon) := \|X'_Y - P_\epsilon X'_Y P_\epsilon\| \le C_Y \epsilon.$$

Reflection takes these middle operators to their adjoints. Consequently

$$|\mu_i^t(\Theta_c Y) - \overline{\mu_i^t(Y)}| \le 4\sqrt{\gamma_{t,i}^2 - 1}\, \|X_{Y,\epsilon}\| + 2 \cdot 2^{|S|/2} \gamma_{t,i} \xi_Y(\epsilon).$$

For fixed support the amplification scales are independent of ϵ. Take $i \to \infty$, then $t \downarrow 0$ with ϵ fixed, then $\epsilon \downarrow 0$.

Proof tasks. Expand in the crossing basis and use the two charge equations to check the temporal phase. Complete the Gaussian exponents to obtain both bounded middle operators. Bound $1 - e^{-\epsilon Q}$ by ϵQ under a Gaussian to prove the norm error. The norm of $X_{Y,\epsilon}$ can grow exponentially in $1/\epsilon$; removing the regularization first is unjustified. Finite sums give general Y.

Source. QED3 pp. 40–42, TeX 3319–3526; `prop:diag-two-layer-principle` and its preceding formulas.

E.a. Exact elimination of a thin fermion layer

INTERFACE. Overview Parent E Target.

Finite algebraic theorem. Use the finite functional and exponential-polynomial class of I.b. Its gauge-covariant hopping coefficients are constant matrices times local link phases, and its local coefficients are integrable against the photon weight. Consider layers such that each site in a layer $\mathbb{L}$ is matched to a unique site in the next layer, its hopping matrix is invertible, and that next site has no other neighbor in $\mathbb{L}$. Assume its remaining neighbors lie in the next two layers. For a local gauge invariant polynomial P, there is a finite linear replacement $R^+_{\mathbb{L}}P$ with no fields in $\mathbb{L}$ and no new fields beyond the next two layers, such that

$$Z_T[\Phi P] = Z_T[\Phi R^+_{\mathbb{L}} P]$$

for every bounded bosonic factor Φ, whenever the enlarged support box embeds. The downward version is identical with the preceding matching. The chamber hopping form in I.b ensures that inverse matching coefficients are constant inverse matrices times inverse link phases; thus replacements stay in the same polynomial class.

Proof tasks. Berezin integration by parts gives, with left derivatives,

$$Z_T[\Phi(\bar\psi\mathcal{D})_\alpha(x')G] = -Z_T[\Phi\partial_{\psi_\alpha(x')}G], \qquad Z_T[\Phi(\mathcal{D}\psi)_\alpha(x')G] = Z_T[\Phi\partial_{\bar\psi_\alpha(x')}G].$$

Solve at the matched vertex x' for the field at $x \in \mathbb{L}$, including its inverse hopping phase. Move that field to the front of each ordered monomial before replacement. The other terms create fields in the next two layers; the derivative terms remove fields. Thus each recursive call removes exactly one field of $\mathbb{L}$. This proves finite termination without a limiting procedure.

Track the charge of the monomial after removing one field; the hopping phase transports that charge to x', so every replacement remains gauge invariant. In the chamber graph the whole thin-layer recursion enlarges the support box by one, regardless of the number of calls. The bosonic factor is unchanged, so the recursion commutes with multiplication by any curvature character.

Source. QED3 pp. 42–43, TeX 3527–3638; exact matching recursions and their support proof.

E.b. A uniform bound for polynomials with arbitrary characters

MODEL-SPECIFIC. Overview Parent E Target.

Theorem. For a fixed local gauge invariant polynomial P and a fixed finite box B_0, there are constants C_P, k_P and a volume threshold such that

$$\sup_{\operatorname{supp}\beta\subset B_0} |\mu_i^t(\varphi_\beta P)| \le C_P \gamma_{t,i}(k_P) \qquad (t \ge 0).$$

The size of the real coefficients of β does not enter the bound. The same conclusion applies to a fixed reflected or transported polynomial. Inputs are E.a, the bounded primed representation in H.b, and the coordinate and amplification bounds N.c, M.b.

Proof tasks. Use exact thin-layer recursions to move all fields to the two crossing layers, allowing the photon exponents to remain in their original box. For each remaining monomial, choose integer paths in the connected two-plane support graph realizing its integer charge. If qj_a is that exponent, its difference from the original real exponent is divergence free. A box-supported cochain filling gives

$$\delta_a = qj_a + d^*\beta_a, \qquad Z_T[\Phi P] = \sum_a b_a Z_T[\Phi \varphi_{\beta_a} Y_a], \quad Y_a \text{ two-layer}.$$

The filling stays inside the fixed box: a partial-sum sweep removes spatial links below its top plane; a second sweep there removes the remaining links, using the divergence and boundary equations.

Represent each $Y_a = G'_{X'_a}$ by H.b. Absorb both curvature characters into the coordinate estimate, then amplify. One may take $C_P = \sum_a |b_a| 2^{|S_a|/2} \|X'_a\|$. All finitely many supports, together with B_0, fit in one fixed box; its radius determines k_P. Reflect the finite identities on the reflected torus before transporting them back; invariance of the diagonal torus is not assumed.

Source. QED3 pp. 43–44, TeX 3639–3688; the support-preserving filling and `eq:diag-polynomial-apriori`.

E.c. The all-flux contour shift and cancellation of an outer plane

MODEL-SPECIFIC. Overview Parent E Target.

Identity. Use the finite all-flux functional of I.b. Let u be a real cochain on finitely many links in one coordinate plane, P a local gauge invariant polynomial, and Φ entire in the curvature variables and bounded on every tube of bounded imaginary curvature. If a fixed enlarged support box embeds in T, then

$$Z_T[\Phi P] = e^{\|du\|^2/2} Z_T[\Phi(F - idu)\varphi_{du}(F) V_u P(\widehat{A} - iu)].$$

Here $V_u = \exp[-S_f(\widehat{A} - iu) + S_f(\widehat{A})]$ is a finite even polynomial, since its exponent is a local nilpotent quadratic Grassmann expression. The norm is the ordinary lattice counting norm.

Proof tasks. Sum the exact integer fluxes generated by the shifted links before shifting. A selected electric plaquette for each link contains no other shifted link; hence d is injective there and this flux action is free. The link phase cells and integer sums unfold to $\mathbb{R}^S$. For fixed other variables each Berezin coefficient is entire on those lines. Selected plaquettes bound it by a product of real Gaussian tails on the finite imaginary strip. Vertical contour sides vanish; absolute integrability permits summing the resulting identities over the remaining flux representatives. Completing the Maxwell square gives exactly the displayed factor and character.

After E.a removes all fields at the lowest plane j, let its remaining spatial exponent be v. Gauge invariance forces the outgoing temporal exponent to be $d_s^* v$. Shift links in plane $j+1$ by $-iu$ with $u_{j+1} = v$. Then

$$(d^* du)_j = -v, \qquad (d^* du)_j^{\perp} = -d_s^* v.$$

Both outer exponents cancel. New links reach at most plane $j+2$; the new polynomial V_u reads only thick layer $j+1$. The reflected operation removes a highest plane. This is the finite reduction used in E.d, not a change of the limiting measure.

Source. QED3 pp. 44–46, TeX 3689–3832; contour identity, strip bound and outer-plane cancellation.

E.d. Iteration with a volume-uniform error after tilting

MODEL-SPECIFIC. Overview Parent E Target.

Reduction theorem. For every fixed polynomial P of I.b and coordinate link plane $c = j_0 + 1/2$, there is a two-layer polynomial Y_P, independent of volume and strength, such that for every fixed lattice isometry g,

$$|\mu_i^t(P^g) - \mu_i^t((Y_P)^g)| \le C_{P,g}t \quad (0 \le t \le 1,\ i \ge i_0(P,g)).$$

The constants may depend on the polynomial, its target plane, the fixed model and the cell array. Inputs are the exact recursions E.a, the contour identity E.c and the character-uniform bound E.b.

Where the error comes from. For a finite curvature shift η, only finitely many reflected copies of O meet its support. Thus $H_\eta(F) = V_i(F - i\eta) - V_i(F)$ is a fixed local trigonometric polynomial in a fixed enlarged box, for all large volumes. Its coefficient norm is bounded by a constant $C(\eta)$, and

$$|e^{tH_\eta} - 1|_1 \le e^{tC(\eta)} - 1 = O(t).$$

Apply the contour identity to $\Phi = e^{tV_i}$. Expand this local error in characters; E.b bounds every coefficient insertion uniformly, so a single outer-plane step costs $O(t)$.

Finite iteration and transport. Eliminate lowest planes until none lies below j_0, then highest planes until none lies above $j_0 + 1$. A highest-plane step can lower its new support only by two, so it preserves the already achieved lower bound. Exact thin-layer recursions then leave only the crossing endpoints. There are finitely many steps and fixed support boxes; take one threshold and sum their errors.

For a reflected polynomial, reflect the exact identity on the reflected torus, then transport it back. A shift η becomes $-\theta^*\eta$. Applying the local tilt-error and polynomial bounds on D_{m_i} gives the transported estimate without assuming that D_{m_i} or its array has that reflection symmetry.

Source. QED3 pp. 46–47, TeX 3833–3941; `thm:diag-polynomial-reduction` and its error proof.

C.a. Ordered weak limits with Gaussian domination and wall RP

MODEL-SPECIFIC. Overview Parent C Target.

Statement and inputs. Let μ_i^t be the bosonic reflected tilts of F.c on an admissible divisible sequence. Assume the local moment bound of M.b. Given any subsequence of volume indices, there is a further common subsequence i_m and positive strengths $t_k \downarrow 0$ with

$$\mu_{i_m}^{t_k} \Rightarrow \nu_{t_k} \quad (m \to \infty) \text{ for each } k, \qquad \nu_{t_k} \Rightarrow \nu_O \quad (k \to \infty).$$

Every such ordered limit has

$$\nu_O(e^{\langle \beta, F \rangle}) \le e^{\|\beta\|^2/2},$$

and retains invariance and RP at the chosen cell walls, initially for bounded continuous local positive-side tests.

Proof tasks. At strengths $1/k$, successively extract convergent volume subsequences using the product tightness from M.b, then diagonalize. The limiting laws are uniformly tight in k, since their one-coordinate exponential moments are bounded with $t_k \le 1$. A strength subsequence therefore converges. The exponential loss $e^{2t_k|O|_1 k_\beta^3}$ tends to one for each fixed support; truncate the exponential to pass the constant-one inequality through the second limit. Here k_β is the fixed geometric amplification scale, not the strength index.

A fixed local side test lies in its finite positive half-period slab for all sufficiently large volumes. Its reflected form is bounded and continuous, so the wall RP and invariance from F.c pass through both weak limits. The whole-side L^2 extension is recorded once in S.a.

The construction extracts laws of curvatures only. No assertion about weak convergence of Grassmann functionals is used.

Source. QED3 pp. 47–48, TeX 3942–3980 and 2687–2714.

C.b. Recovering the missing coordinate reflections

MODEL-SPECIFIC. Overview Parent C Target.

Theorem. For the reflected tilts on any admissible divisible sequence, every coordinate link reflection θ_c, $c \in \mathbb{Z} + 1/2$, and finitely supported real β satisfy

$$\lim_{t \downarrow 0} \limsup_{i \to \infty} |\mu_i^t(\varphi_\beta \circ \theta_c^*) - \mu_i^t(\varphi_\beta)| = 0.$$

Consequently every ordered comparison state of C.a is invariant under all these reflections and under $2\mathbb{Z}^3$ translations. Inputs are the transported reduction of E.d and the two-layer reflection estimate of H.b.

Proof tasks. Represent the curvature character by $P(\widehat{A}) = e^{i\langle d^*\beta, \widehat{A}\rangle}$ on a local lift. Reduce P at the chosen plane to Y_P, and use the transported estimate for its reflection. The two reductions cost $O(t)$. The remaining difference is the two-layer error of H.b. Take volume to infinity, then strength to zero with regularization fixed, then remove regularization. Only at this point pass character expectations through the bosonic weak limits. This gives

$$\nu_O(\Theta_c \varphi_\beta) = \overline{\nu_O(\varphi_\beta)}.$$

Since $\Theta_c \varphi_\beta = \varphi_{-\theta_c^* \beta}$, and complex conjugation of a character expectation changes β to $-\beta$, this is reflection invariance of all characteristic functions. Uniqueness of finite-dimensional Fourier transforms and cylinder generation of the Borel sigma-algebra give invariance of the law. The product of reflections at c and $c+1$ is translation by $2e_j$. Repeat in each coordinate direction.

Scope of the conclusion. The finite tilted laws were never assumed invariant under these link reflections. The conclusion supplies exactly the even translations used to move an observable into a cell with a prescribed diagonal wall.

Source. QED3 p. 48, TeX 3990–4041; `eq:diag-character-reflection` and `eq:diag-even-translation`.

V.a. The unequal-height slab estimate

MODEL-SPECIFIC. Overview Parent V Target.

Theorem. Use the adapted slab kernels of F.a. Cut a torus of fixed cross-section into $2k$ consecutive admissible slabs with possibly different heights h_j, each large enough for its local integrand. For slab j, let T_j have this cross-section and straight normal period $2kh_j$, with antiperiodic normal character. Then

$$|Z_T[\prod_{j=1}^{2k} G_j]| \le \prod_{j=1}^{2k} Z_{T_j}[\mathrm{fold}_{T_j} G_j]^{1/(2k)}, \qquad |T| = \sum_{j=1}^{2k} \frac{|T_j|}{2k}.$$

The folded factors are nonnegative. The tori and embedding conditions must be admissible individually; slab heights need not agree.

Proof tasks. The exact cyclic trace has a Hilbert–Schmidt operator for each slab and a unitary closing identification. Apply Schatten Hölder with the common exponent $2k$. The quantity $\mathrm{Tr}((\mathfrak{S}[G_j]^*\mathfrak{S}[G_j])^k) = \|\mathfrak{S}[G_j]\|_{2k}^{2k}$ is its repeated reflected fold, on its own torus T_j, rather than on one common-height torus. Its cross-sectional area is fixed, so summing heights gives the displayed volume identity. Track characters and the local embedding conditions in each repeated fold as in the equal-height case. Each trace is taken at its own starting boundary, as in F.a.

Use. Take $2k-1$ regular slabs of width h containing the chosen copies of O, and one empty wider slab. The empty slab may produce a much larger exceptional torus. Its contribution carries the Hölder weight $1/(2k)$; bounding its raw volume instead of its weighted volume loses the small-error estimate. V.b tracks those weights through three successive folds.

Source. LPM1, Proposition 10.1, using the corrected Proposition 4.2. S2 `prop:unequal-slabs`, TeX 33984–34207; one-step weighted fold `lem:one-step`, 34419–34483. Estimated full proof is included in the parent budget.

V.b. The arbitrary-side comparison with its exact error

MODEL-SPECIFIC. Overview Parent V Target.

Statement. Use fixed parameters, positive normalizations, the coordinate pressure limit f_∞ of F.b and the unequal fold V.a. Let O lie inside a diagonal cell of even width $h \geq 18$, with support radius $r \geq 1$ and $h > 8r + 8$. For $4h \mid m$, $m \geq 4h$ and $n \geq m$, define

$$V_{\mathrm{reg}} = \frac{(m-h)^2(2m-h)}{2}, \quad V_{\mathrm{exc}} = n^3 - V_{\mathrm{reg}},$$

$$\Delta'(n,m,h) = f(m/2, 2m, m) - f_\infty + \frac{V_{\mathrm{exc}}}{V_{\mathrm{reg}}}(\bar{f} - f_\infty),$$

where $\bar{f}$ is the common upper pressure bound specified below. For the full reflected tilt on D_m,

$$\mu_{C(n)}(O) \leq \mu^t_{D_m}(O) + \frac{|Q|\Delta'(n,m,h)}{t} \qquad (t > 0).$$

There is no parity restriction on n.

Proof tasks. Perform the unequal folds successively along the diagonal frame:

$$C(n) \to D'(m/2, n, n) \to D(m/2, m, n) \to D_m.$$

At step a, use $2k_a - 1$ regular slabs of width h and one empty slab of height $p_a - (2k_a - 1)h$. Its exceptional torus T^{exc}_a has the preceding cross-section and normal period $2k_a[p_a - (2k_a - 1)h]$. The heights and accumulated Hölder weights are

a	p_a	k_a	ϑ_a
1	n	$m/(2h)$	$1/(2k_1)$
2	$2n$	m/h	$(2k_1-1)/(4k_1k_2)$
3	n	$m/(2h)$	$(2k_1-1)(2k_2-1)/(8k_1k_2k_3)$

There are $R = \prod_a (2k_a - 1)$ regular factors, each with exponent $1/(8k_1k_2k_3)$. Since $|Q| = m^3/(8k_1k_2k_3)$, their weighted volume is $R|Q| = V_{\mathrm{reg}}$. Substituting the three one-step volume identities of V.a yields $n^3 = \sum_a \vartheta_a |T^{\mathrm{exc}}_a| + V_{\mathrm{reg}}$, so the exceptional weighted sum is exactly V_{exc}. Bound these pressures by $\bar{f}$ and the original cubic pressure from below by f_∞. Jensen for translated copies and convexity of the diagonal log partition function give the comparison after dividing by the regular copy count.

Here $\bar{f} = 192 \log H_M + 3 \log(r_q C_q)$, $H_M = |M| + (1+\rho)\sum_i w_i/2$ and $C_q = \sup_y \sum_{l \in \mathbb{Z}} e^{-(y + r_q l)^2/2}$; the finite determinant and Gaussian estimates give this bound on every exceptional torus.

Source. S2 `prop:fold-any-side`, TeX 34517–34982, with pressure and one-step inputs in 33813–34516. This statement uses only its original-law case.

V.c. Divisible companions and the ordered local inequality

MODEL-SPECIFIC. Overview Parent V Target.

General arithmetic lemma. For any integer sequence $n_i \to \infty$, there are integers m_i with $m_i \le n_i$ eventually, $m_i/n_i \to 1$, and every positive integer dividing m_i eventually. For $n \ge 64$ one explicit choice is

$$k(n) = \max\{k \ge 1 : 8k! \le \sqrt{n}\}, \qquad m(n) = 8k(n)! \left\lfloor \frac{n}{8k(n)!} \right\rfloor .$$

Then $0 \le n - m(n) < \sqrt{n}$ and $k(n) \to \infty$. Define the finitely many smaller terms arbitrarily, for example as 8.

Application. For each fixed cell width h, V.b applies to (n_i, m_i) eventually. Its error tends to zero because all sides of $C(m_i/2, 2m_i, m_i)$ grow and

$$\frac{V_{\mathrm{reg},i}}{n_i^3} = \left(\frac{m_i}{n_i}\right)^3 \left(1 - \frac{h}{m_i}\right)^2 \left(1 - \frac{h}{2m_i}\right) \to 1.$$

Thus $\Delta_i' \to 0$. The companions meet every eventual divisibility condition needed in M–C.

Suppose the original $\mu_{C(n_i)}$ converge to ν on a prescribed subsequence, and form a comparison state ν_O on its companion indices as in C.a. At each fixed $t_k > 0$, take the volume limit in V.b; its error vanishes. Then take $t_k \downarrow 0$. Bounded continuity of O gives $\nu(O) \le \nu_O(O)$. This conclusion compares an expectation of the original limit to a comparison state. It does not identify original and companion laws.

Source. S2 `lem:companions-exist`, TeX 35069–35130; `thm:comparison-any-side`, 35175–35277. The factorial choice and limit passage are both supplied there.

S.a. A compact convex RP class and the full side sigma-algebra

INTERFACE. Overview Parent S Target.

Statement. On the Polish closed-cochain space Ω of I.a, fix a lattice mirror Π and its tensor pullback θ^*. Let $\mathcal{R}(\Pi)$ consist of probability laws λ such that

$$\lambda(e^{\langle\beta,F\rangle}) \le e^{\|\beta\|^2/2} \quad (\beta \text{ real, finitely supported}),$$

which are reflection invariant and satisfy $\lambda((\Theta A)A) \ge 0$ for all bounded continuous local positive-side A. Then $\mathcal{R}(\Pi)$ is compact and convex in the weak topology. Each member is RP on all of $L^2(\Sigma_+, \lambda)$.

Proof tasks. The Gaussian estimate gives a uniform one-plaquette tail $2e^{-R^2/2}$. Choose coordinate radii with summable tails to obtain uniform tightness on the countable product, intersected with $dF = 0$. Truncation of exponential integrals passes their bound to weak limits. Continuous pullback passes reflection invariance; bounded continuous local reflection forms pass positivity. These conditions are affine equalities or affine inequalities in the law, proving convexity.

For a finite set of side coordinates, bounded continuous functions are dense in the marginal L^2 space. Since reflection invariance makes Θ an L^2 isometry, the reflected-form difference is at most

$$\|A - A_l\|_2(\|A\|_2 + \|A_l\|_2).$$

This gives Borel local tests. Enumerate all positive-side plaquettes; conditional expectations on their increasing finite sigma-algebras converge in L^2. Apply the same estimate to obtain the entire side. The assertion also covers the opposite open side by reflection.

Use. Take $\Pi = \{x_1 - x_2 = c_0\}$. C.a puts each comparison state in this class. S.b will put the original cubic limit there.

Source. QED3 pp. 33, 48–49, TeX 2687–2714 and 4043–4059; whole-side density also appears in the lattice-limit argument of S2, TeX 35419–35553.

S.b. A model-independent convex separation theorem

INTERFACE. Overview Parent S Target.

Theorem. Let X be a Polish space, and let $\mathcal{A} \subset C_b(X;\mathbb{R})$ be a real vector space determining finite signed Borel measures. Let $\mathcal{R}$ be a nonempty convex weakly compact set of probability laws on X, and let ν be another probability law. If

$$\text{for every } O \in \mathcal{A} \text{ there is } \lambda_O \in \mathcal{R} \text{ with } \nu(O) \le \lambda_O(O),$$

then $\nu \in \mathcal{R}$. No common choice of λ_O is required.

Proof. Give finite signed measures the locally convex topology generated by integration against $\mathcal{A}$. The determining hypothesis makes it Hausdorff. A continuous linear functional depends on finitely many of these integrals, so is integration against their real linear combination, again an element of $\mathcal{A}$. Since these tests are bounded and continuous, weak compactness of $\mathcal{R}$ makes it compact in this topology. If $\nu \notin \mathcal{R}$, strict separation gives $\nu(O) > \max_{\lambda\in\mathcal{R}} \lambda(O)$ for some $O \in \mathcal{A}$, contrary to the hypothesis.

Lattice specialization. Take $X = \Omega$ and the real local trigonometric polynomials from I.a. They determine signed measures: Fourier uniqueness determines each finite-dimensional marginal, and cylinder sets determine a Borel measure on the countable product. Take the compact convex RP class of S.a. The remaining model input is precisely the displayed domination for every real polynomial; S.c supplies it.

Source. QED3 pp. 48–49, TeX 4097–4128; `lem:diag-separation`. This is the general formulation of its supplied separation argument.

S.c. Position the observable, dominate it, and prove all nine families

MODEL-SPECIFIC. Overview Parent S Target.

Inputs and conclusion. Let ν be any original cubic weak limit, with the inherited translation and cubic symmetries of I.a. Use comparison states from C, the compact RP class of S.a and separation S.b. For divisible original sides use F.b; for arbitrary original sides use V.c. Then ν is RP at every diagonal lattice mirror and hence satisfies A.

Place one arbitrary real test. Fix $c_0 \in \mathbb{Z}$ and a real local trigonometric polynomial O in $y_0 + [-r, r]^3$, $r \geq 1$. Choose an admissible even width $h > 8r + 8$. There is an even integer a with

$$c_0 + 2r < \ell_-(y_0) + a < c_0 + h - 2r.$$

Put $v = ae_1$. Choose the other two offsets so the whole translated box lies in the open diagonal cell whose lower ℓ_- wall is c_0. This uses only intervals of length greater than one; h is fixed before the volume and strength limits.

Dominate the test in its original position. Tilt using O^{τ_v} and construct a comparison state ν_O subordinate to the original convergent index subsequence. The finite comparison, followed by volume then strength limits, gives

$$\nu(O^{\tau_v}) \leq \nu_O(O^{\tau_v}).$$

The original law is invariant under all lattice translations; C.b makes ν_O invariant under this particular even v. Consequently $\nu(O) \leq \nu_O(O)$, with $\nu_O \in \mathcal{R}(\{x_1 - x_2 = c_0\})$. This holds for every O in the real determining space. Separation puts ν in the same RP class.

Repeat for each c_0. A signed coordinate permutation conjugates this reflection and its side to any $x_j \pm x_k = c_0$; cubic invariance identifies their reflection forms. S.a gives the full side L^2 statement. The coordinate families were already supplied by I.a.

Source. QED3 pp. 48–49, TeX 4061–4095 and 4130–4189; arbitrary-side domination and assembly: S2 TeX 35280–35553.